Chapter 18, Problem 35P

Problem

A signal $f(t)$ has Fourier transform $F(\omega) = \dfrac{1}{2 + j\omega}$

Determine the Fourier transform of the following signals:

(a) $x(t) = f(3t - 1)$

(b) $y(t) = f(t)\cos 5t$

(c) $z(t) = \dfrac{d}{dt} f(t)$

(d) $h(t) = f(t) * f(t)$

(e) $i(t) = t\, f(t)$

Step-by-step solution

Step 1 of 5

(A) $x(t) = f\left[3\left(t - \frac{1}{3}\right)\right]$

Using the scaling + time shifting properties

$$X(\omega) = \frac{1}{3}\cdot\frac{1}{2 + \frac{j\omega}{3}} e^{\frac{-j\omega}{3}} = \frac{e^{\frac{-j\omega}{3}}}{(6 + j\omega)}$$

Step 2 of 5

(B) Using the Modulation property

$$Y(\omega) = \frac{1}{2}\left[F(\omega + 5)(\omega - 5)\right]$$

$$= \frac{1}{2}\left[\frac{1}{2 + j(\omega + 5)} + \frac{1}{2 + j(\omega - 5)}\right] = \frac{1}{2}\left[\frac{1}{j\omega + 7} + \frac{1}{j\omega - 3}\right]$$

Step 3 of 5

(C) $Z(\omega) = j\omega F(\omega) = \dfrac{j\omega}{2 + j\omega}$

Step 4 of 5

(D) $H(\omega) = F(\omega)F(\omega) = \dfrac{j.1}{(2 + j\omega)^2}$

Step 5 of 5

(E) $I(\omega) = j\frac{d}{d\omega}F(\omega) = \frac{j(0-j)}{(2+j\omega)^2} = \frac{1}{(2+j\omega)^2}$